

ADVANCING FIRST CLASS PRE-K IN ALABAMA

Same Gains, Different Levels: What GOLD Scores Say about Growth and Gaps

Classroom-level kindergarten-readiness assessment patterns, 2021–22 to 2023–24

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Do children in First Class Pre-K's poorest classrooms gain as much across the school year as children in its most affluent? We bring the program's own child assessment records—Teaching Strategies GOLD, which teachers complete each fall and spring—to that question for the first time in this series, aggregating 68,888 student records into classroom-year cells (4,194 in literacy) across three school years. We find two patterns that belong together. Classroom poverty is associated with substantially lower assessed skills at the end of the year, by about 20 scale points in literacy when an all-poverty classroom is compared with a no-poverty one. Yet classroom poverty is essentially unrelated to fall-to-spring *gains* in every one of the six developmental domains: classrooms where at least three-quarters of students live in poverty gained 50.9 literacy points on average, while classrooms where at most a quarter do gained 51.3.

Our analysis yields three findings:

- **The level gap is already present in the fall.** Spring literacy scores fall about 20 points per 100 points of classroom poverty share, and fall scores show nearly the same gradient of 16 points, which suggests that most of the spring gap was present at entry (figure 1).

- **Growth is flat across the poverty gradient.** The fall-to-spring gain moves by only -4 to $+4$ points per 100 points of poverty share across the six domains, a small fraction of the 11-to-28-point level gaps (figure 2).
- **The pattern repeats in each year we observe.** In literacy, the level gradient runs from -19 to -22 in each of the three years, while the gain gradient stays between -3 and -5 .

Two cautions frame everything that follows. Because these are descriptive associations in teacher-administered assessment data, with no comparison group of children outside pre-K, they cannot say what the program *causes*. And equal gains are not the same as closed gaps, because children who enter behind and grow in parallel with their peers remain behind.

The Equity Question and the GOLD Assessment

Alabama's pre-K case has rested largely on quality standards and national rankings, which are measures of inputs and reputation rather than of children. The program also collects a direct and universal measure of children's development: GOLD, an observational assessment that teachers complete for every enrolled child in fall and spring across the literacy, mathematics, language, cognitive, social-emotional, and physical domains. This series' national briefs established where Alabama stands among the states, but they could say little about what happens inside the school year. We take that question up here, asking whether classroom poverty is associated with how much children gain between fall and spring as well as with where they end up. That is the equity question a mixed-delivery, statewide program must be able to answer.

This brief, the fourteenth in the *Advancing First Class Pre-K in Alabama* series, is the first to bring the program's child assessment records into the series' evidence base.

BOX 1

Data and Methods

Data. First Class Pre-K student assessment records, school years 2021–22 through 2023–24: 68,888 student-year records with GOLD scale scores. We aggregate to classroom-year cells (4,194 cells containing at least 5 assessed students; the literacy cells cover 62,680 students). Classroom poverty is the share of the classroom's students flagged as living in poverty in program records.

Measures. GOLD is an observational assessment completed by teachers; scale scores are Teaching Strategies' interval metric. We define "level" as the classroom's mean spring score and "growth" as the mean fall-to-spring gain. Gradients are weighted least-squares slopes across classrooms, expressed per 100 percentage points of classroom poverty (that is, comparing a hypothetical all-poverty classroom with a no-poverty one).

What this is not. We make no causal claims: there is no comparison group, teachers assess their own students, and classroom poverty travels with many other classroom characteristics. Student-level records are restricted, and we report nothing below the classroom grain.

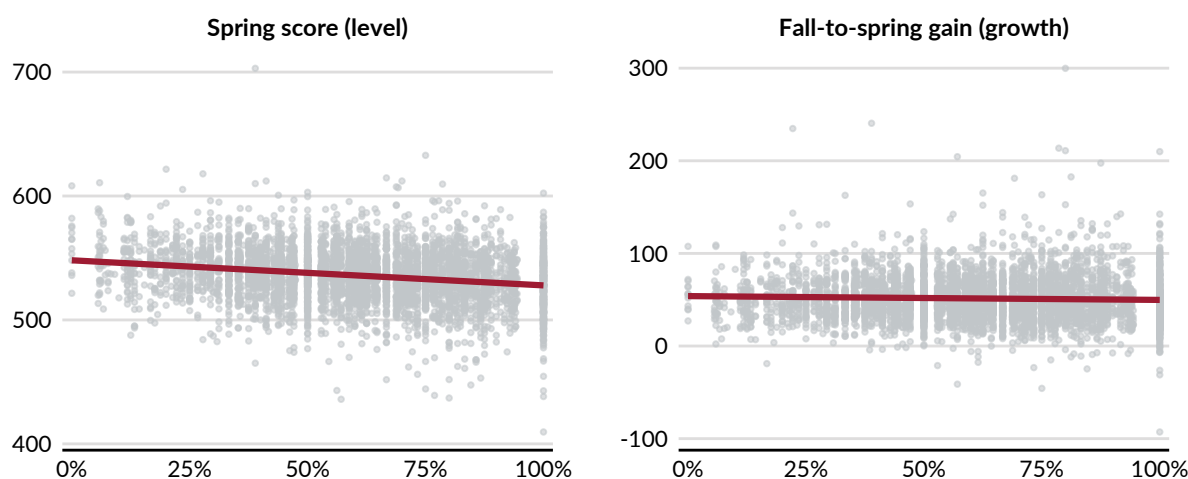
Literacy Levels and Literacy Growth across Classroom Poverty

Figure 1 plots every literacy classroom-cell twice, first by its spring level and then by its fall-to-spring growth, each against the classroom's poverty share. In the left panel, higher classroom poverty is associated with visibly lower spring scores, by about 20 scale points across the full poverty range on a spring mean of about 535. In the right panel that association all but disappears, because classrooms average 51.5 points of literacy growth almost regardless of which children are enrolled.

FIGURE 1

Spring Levels Fall with Classroom Poverty; Gains Do Not

Literacy: classroom mean spring score and mean fall-to-spring gain, by classroom poverty share, 2021–22 to 2023–24



Source: Authors' analysis of ADECE First Class Pre-K GOLD assessment records, 2021–22 to 2023–24.

Notes: ADECE = Alabama Department of Early Childhood Education. Each point is a classroom-year cell (4,194 cells; at least 5 assessed students each—39 smaller cells covering 89 students are excluded); lines are weighted least-squares fits. Axis scales differ by panel.

The fall gradient completes the interpretation, because entering literacy scores already slope -16 points across the poverty range. The spring gap of -20 we observe is therefore likely to be, overwhelmingly, the entry gap carried forward, accompanied by a small additional widening of -4 points of gain per 100 points of poverty rather than by any catch-up.

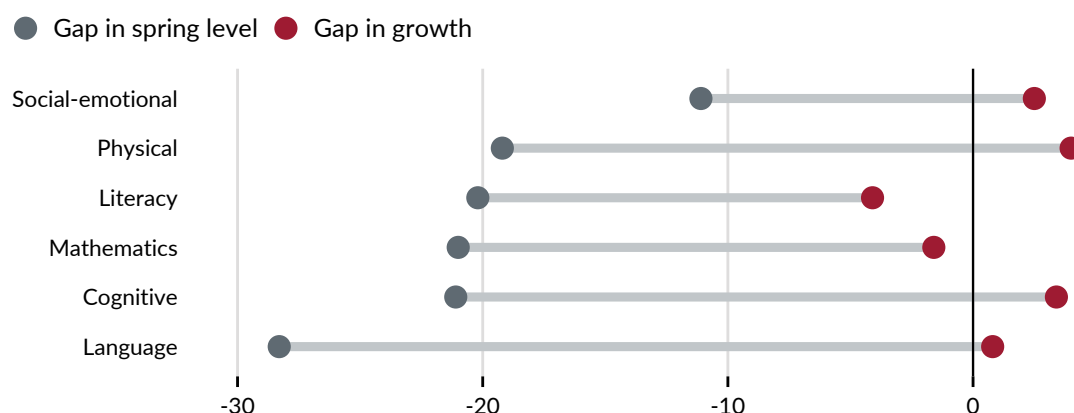
Level and Growth Gradients across the Six GOLD Domains

Literacy is not a special case (figure 2). Across the six GOLD domains, higher classroom poverty is associated with spring levels lower by 11 points in social-emotional development and by as much as 28 points in language per 100 points of classroom poverty, while the growth gradients cluster tightly around zero, running from -4 in literacy to $+4$ in physical development. In four of the six domains the growth gradient is zero or slightly *positive*, meaning that high-poverty classrooms gain marginally more.

FIGURE 2

Level Gaps Are Large in Every Domain; Growth Gaps Hug Zero

Slope per 100 percentage points of classroom poverty: spring level versus fall-to-spring growth, six GOLD domains



Source: Authors' analysis of ADECE First Class Pre-K GOLD assessment records, 2021–22 to 2023–24.

Notes: Slopes from weighted least-squares fits across classroom-year cells, pooled over three school years. Negative values mean lower scores (or smaller gains) in higher-poverty classrooms.

The two gradients suggest something specific: whatever mix of teaching, curriculum, and context accompanies the growth we observe, it may be present at similar strength across the program's economic spectrum. The equity problem visible in these data is therefore not unequal growth but the fact that equal growth does not close unequal starting points.

Discussion

Our results speak to the program's equity record in two directions at once, and both deserve to be stated. A statewide, mixed-delivery program in which assessed growth is indistinguishable between its poorest and its most affluent classrooms is not the default result, and the pattern is at least consistent with

a quality model that reaches classrooms across the economic spectrum. Although our data cannot establish that the program produced the growth we record, the absence of a growth gradient is the affirmative half of the story and may deserve equal billing with the gap in levels.

The second half of the finding points before age four. Classroom poverty is associated with where children start far more than with how fast they grow, and that ordering redirects attention to the years before enrollment. If the spring gap is mostly the fall gap, then closing it within the pre-K year alone would require high-poverty classrooms to *outgrow* affluent ones substantially, which no large program reliably does. Our findings may therefore redirect the equity conversation toward birth-to-three programs, family supports, and the earlier learning environments that this series' access briefs map geographically.

Targeted enrichment is the testable middle path. A supplemental dose—extended day, literacy specialists, summer programming—aimed at high-poverty classrooms would be an attempt to bend the one line these data show to be flat. Were such a strategy adopted, it would likely need to be built for evaluation from the outset, because the data in hand cannot distinguish a program that cannot accelerate growth from one that has never specifically tried.

Limitations

These analyses have several limitations. Most fundamentally, we have no counterfactual: without a comparison group, the growth we observe cannot be attributed to the program, since four-year-olds also grow outside pre-K. Importantly, the scores are teacher-administered: GOLD is observational and teachers rate their own students, so rating practices may vary with classroom context and the score gaps reported here likely reflect measurement as well as skill. Classroom poverty is also only one lens, given that the poverty flag is a binary program indicator aggregated to classrooms that proxies, imperfectly, a bundle of family and community differences. Finally, equal growth is not equal outcomes: nothing here says gaps close, and the data show instead that they persist at about the size children arrive with. These limitations bound the claim we make, which is that children grow at about the same measured rate in every classroom the program serves and that the readiness gap remaining at the end of the year is the one that was present when they arrived.

References

Lee, JoonHo, and Alison Hooper. 2026. "A Decade of Perfect Tens: Where Does Alabama Pre-K Stand in the Nation?" Advancing First Class Pre-K in Alabama, Brief 4. Tuscaloosa: University of Alabama, College of Education.

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ABOUT THIS PROJECT

The Advancing First Class Pre-K in Alabama project examines First Class Pre-K funding, implementation, access, and comparisons with other state pre-K programs. Its goal is to offer recommendations for expanding First Class Pre-K while maintaining high levels of quality.

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